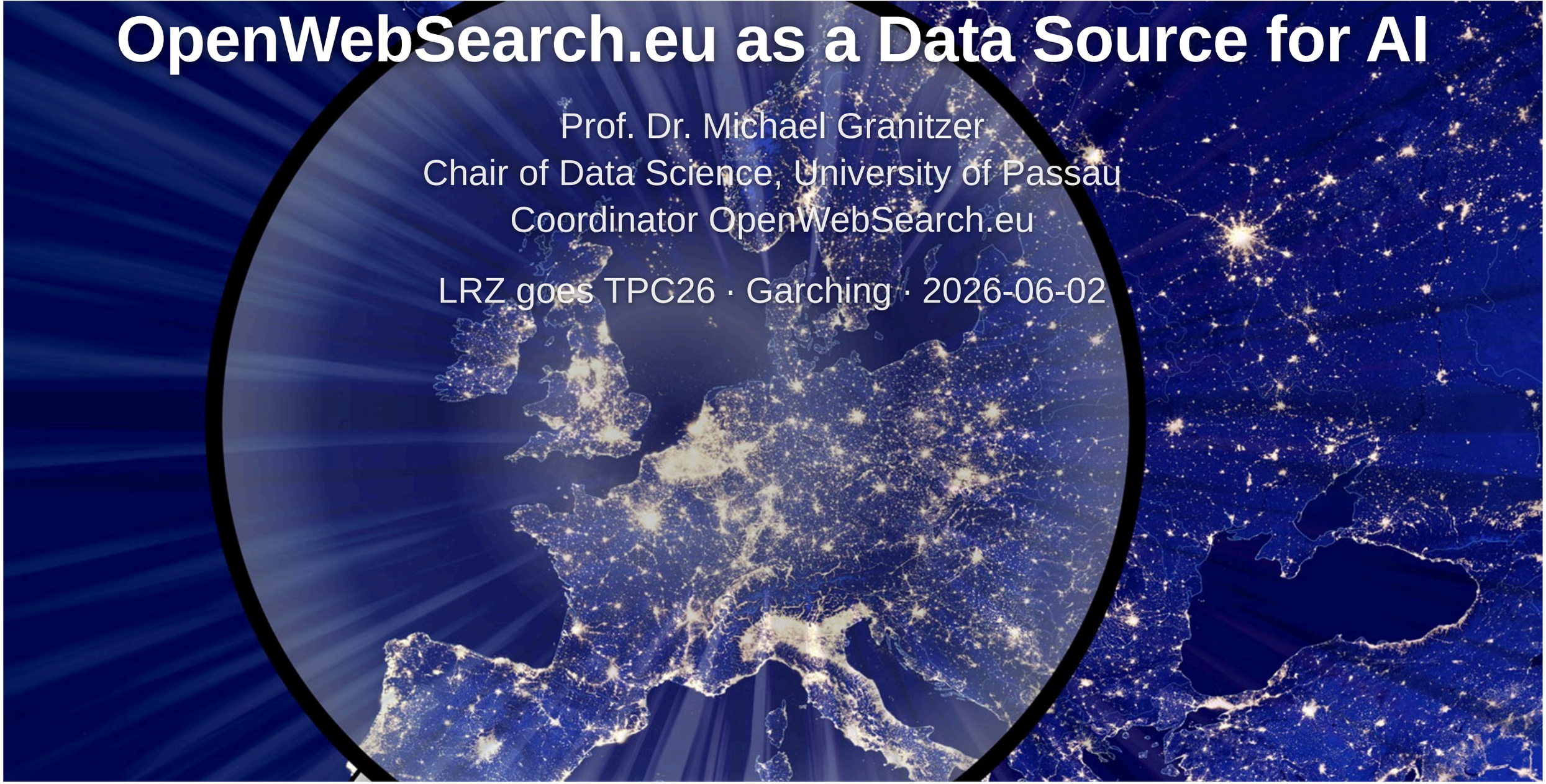
OpenWebSearch.eu as a Data Source for AI

Prof. Dr. Michael Granitzer
Chair of Data Science, University of Passau
Coordinator OpenWebSearch.eu

LRZ goes TPC26 · Garching · 2026-06-02



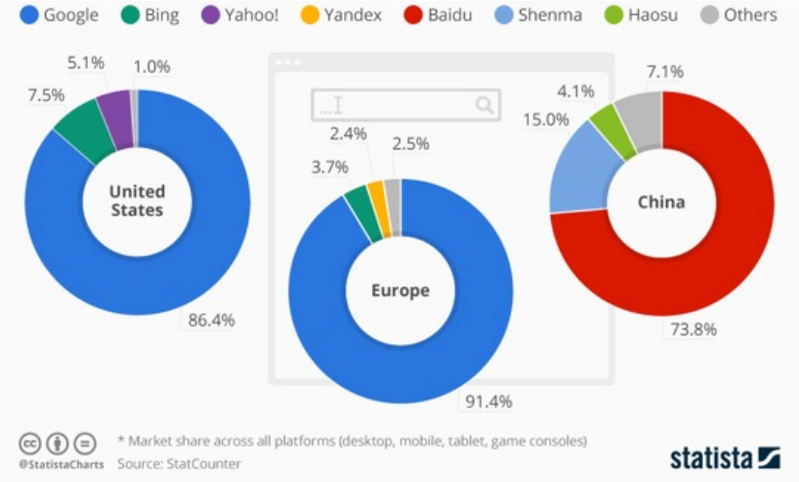
Funded by

Bundesministerium
für Forschung, Technologie
und Raumfahrt

Bayerisches Staatsministerium für
Wirtschaft, Landesentwicklung und Energie

Web Search & Web Data - a Critical Infrastructure

- A **critical infrastructure** for society — and now for AI — comparable to satellite navigation
- A **market oligopoly**: controlled by a handful of gatekeepers (Google, Microsoft, Baidu, …)



Consequences

- **Reduced user choice**
- **Dependence in the information space**
- **Smaller researches and innnovators are locked out**

> How to enable researchers, innovators and smaller companies to utilise the web as **data resource** for training and fine-tuning AI systems and to develop **AI-based Search Systems** without massive investments in infrastructure?

OpenWebSearch.eu

Building an **Open Index of the Web** to lower the cost for researchers, innovators, and companies to **tap the Web as ressource**

Four objectives

1. **Open technology stack** — open pipelines for crawling, preprocessing, indexing
2. **Resource provision** — a network of providers on EuroHPC / AI Factories
3. **Added-value services** — dashboard, tools, data-access APIs
4. **Bootstrapping the ecosystem** — third-party calls, community, use-cases in **search, analytics, and AI**.

14 core partners



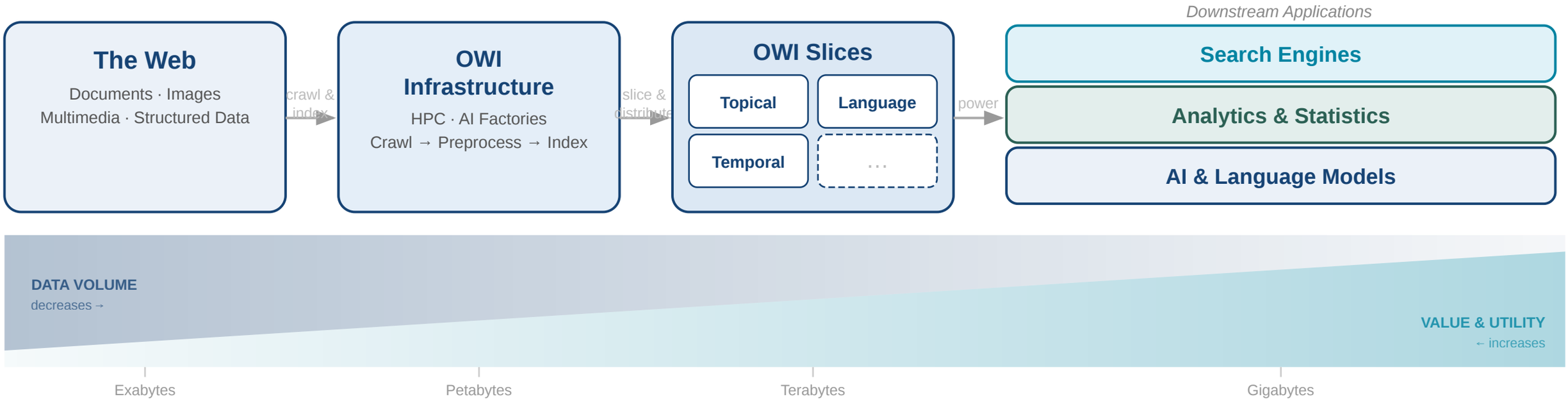
Uni Passau · LRZ · CERN · CSC · IT4I · DLR · TU Graz · Bauhaus-Uni Weimar · Radboud · Uni Leipzig · A1 Telekom · Open Search Foundation · NLnet · SUMA-EV — plus 16 funded third-party projects.

Funded by Horizon Europe (Next Generation Internet programme).

The Goal: An Open Web Index

A **web index** is the data structure behind every search engine — fast, query-based access and ranking over web documents.

> **Our proposition:** a collaboratively created, **open and transparent** web index, running on Europe's HPC centres and AI Factories.



Building the Index

Federated HPC Workflow

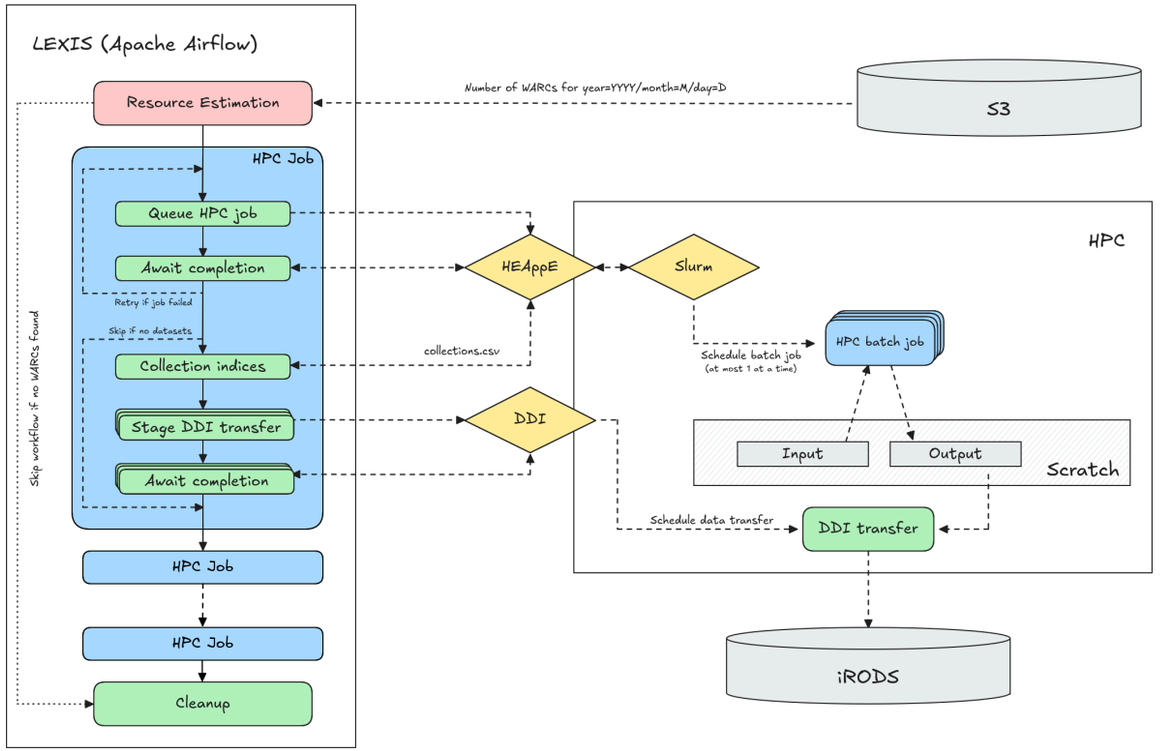
- Continuous Crawling on Cloud Partitions
- Daily Batch HPC Workflows: WARC processing · indexing · semantic enrichment, running
- Runs federated at **LRZ** (Garching) · **IT4I** (Ostrava) · **CSC** (Helsinki)

Federated storage

- Object Storage for Crawl Data
- File Storage for Index Provision
- SQL and slow full-text search for file storage based slice and dice

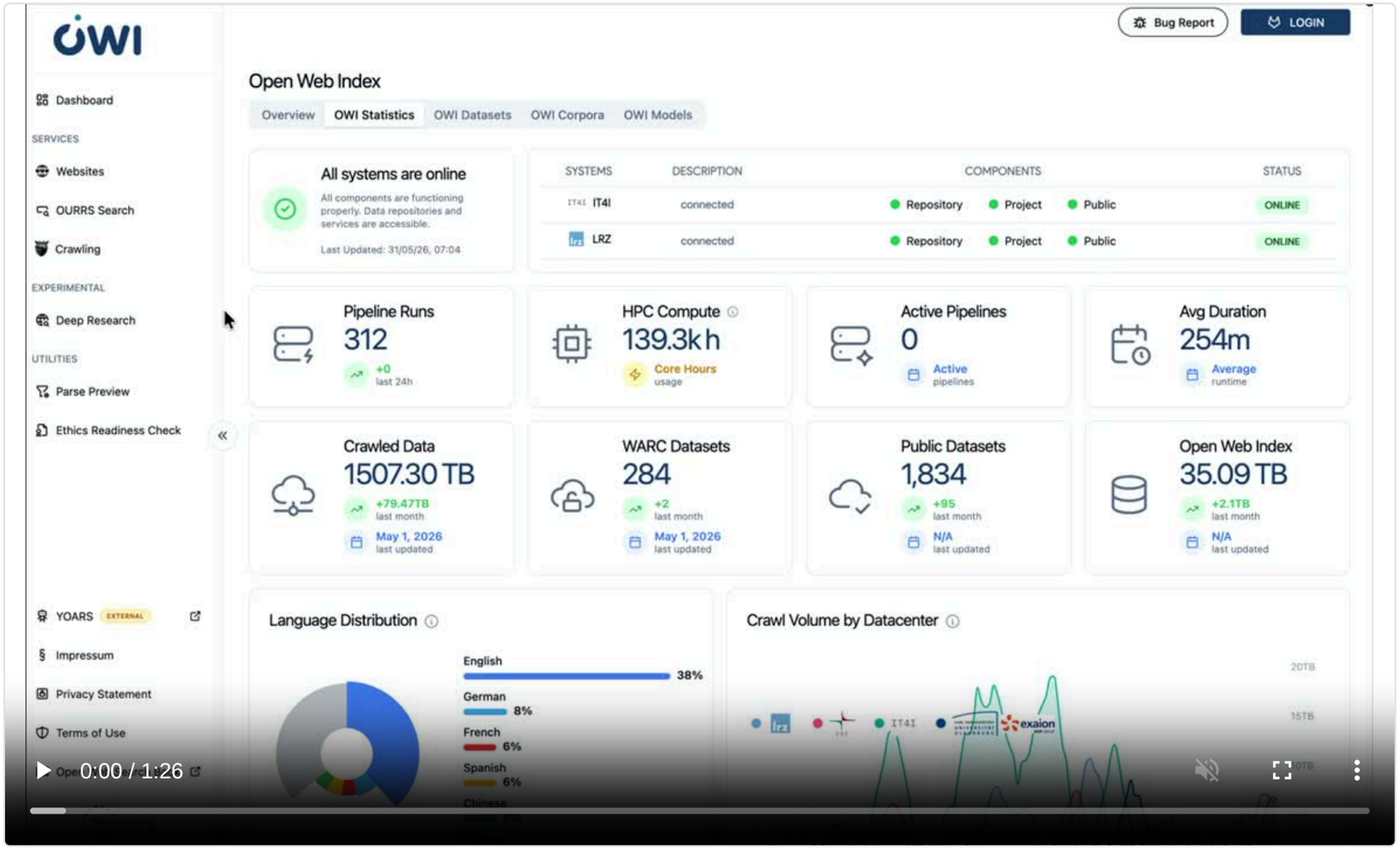
Orchestration

- LEXIS (Apache Airflow) orchestrates jobs across schedulers
- stages data transfers, aligns daily output across all centers



LEXIS / Apache Airflow: federated OWI pipeline orchestration

The Open Web Index Today



▶ Statistics 0:10 ▶ Datasets 0:35 ▶ Corpora 0:51Inin ▶ Models 1:14

OWI dashboard — live operational statistics. http://openwebindex.eu

1,507 TB
data crawled

35 TB
Open Web Index

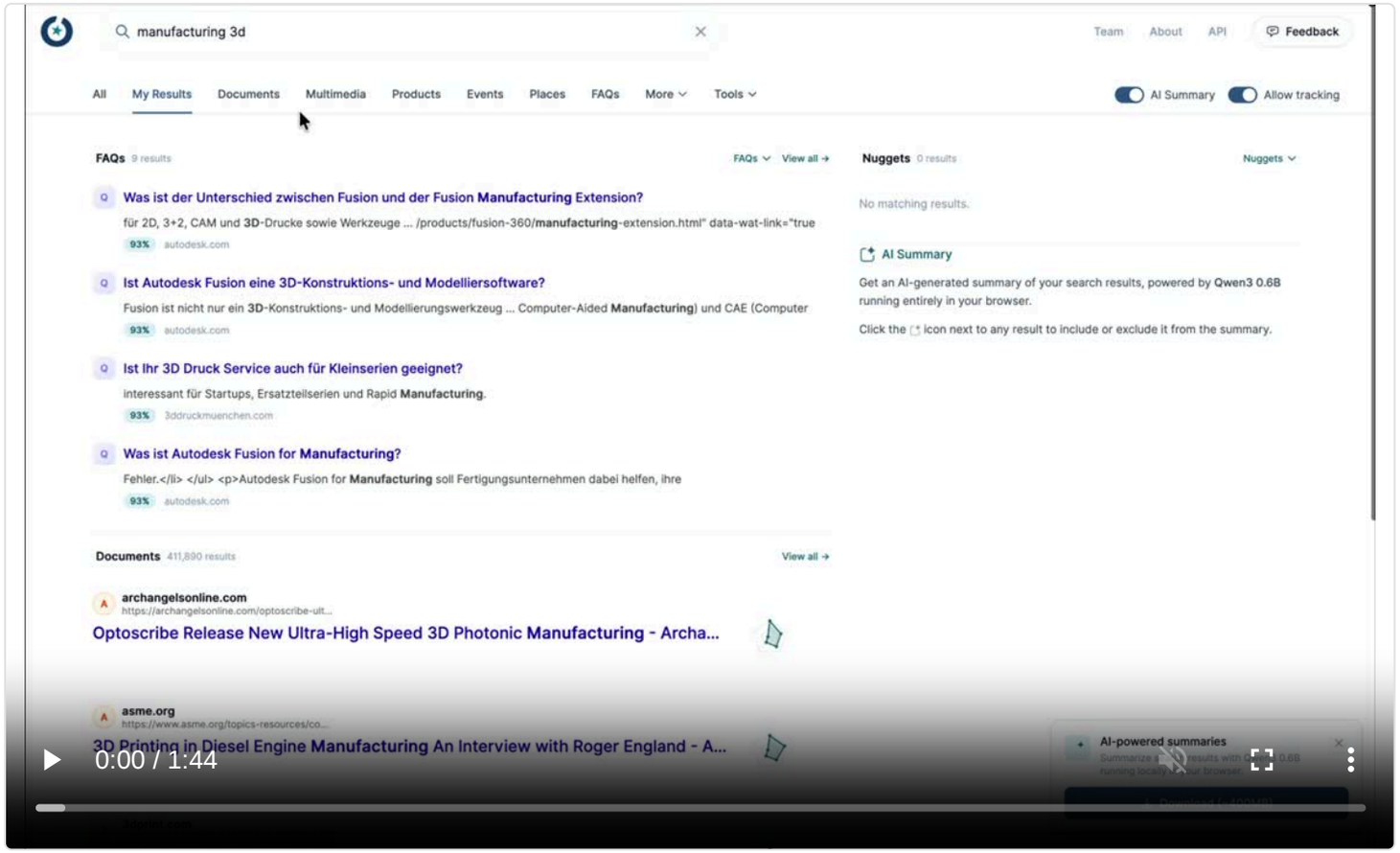
1,834
public datasets

284
WARC datasets

Continuously refreshed — **+79 TB crawled and +95 datasets in the last month** alone.

OWI dashboard, figures as of May 2026.

OUR^2S Web Search Engine



2 Months of data
04/25 + 05/25

598.1M Pages

453.4M Entities

14 verticals
documents · media · products · events · places · FAQs · Scholarly · Artefacts · Actions · Pages · Sites · Persons · Conversations · Courses

in-browser AI
private Qwen3 summaries, ~400 MB

OUR²S turns the OWI into a **working search engine** — documents *and* structured entities, summarised privately in the browser.

OUR²S (ourrs) live demo; search index snapshot April/May 2025.

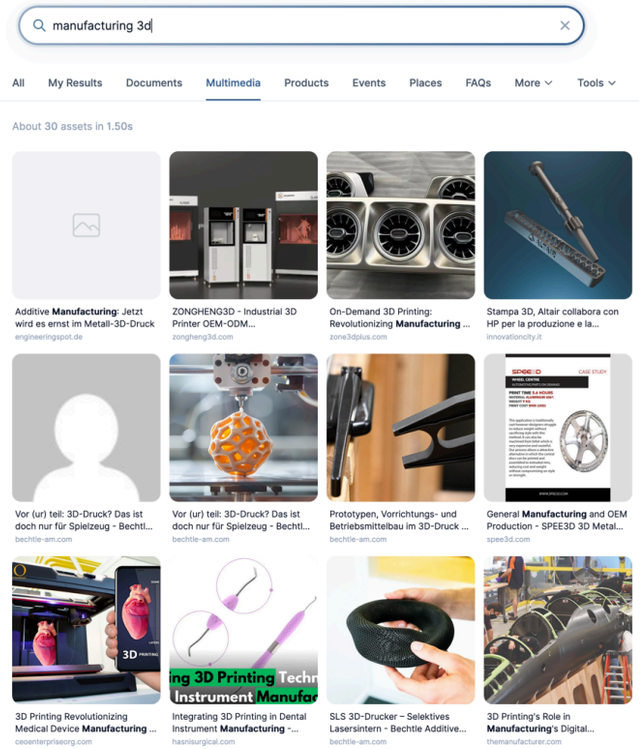
Web data for training & fine-tuning

Modalities: The Index as a Multimodal Filter

MEDIA ASSETS IN ONE MONTH OF THE OPEN WEB INDEX

600M web documents	**142.1M** media assets	**127.8M** images
4.9M videos	**532k** other media	**208k** audio

The index lets you **select training data** by modality, language, quality, and host — *before* paying to download and process it. Storage costs drop when re-downloading text & multimedia can be **targeted and selective**.



Example: multimedia results for a manufacturing query — images, 3D models, product shots.

Asset counts by schema.org type from the Open Web Index API (api.ourrs), one-month corpus, queried 2026-05-30.

Domain-Specific Data

"MANUFACTURING" DOCUMENTS IN A SINGLE MONTH OF THE OPEN WEB INDEX

6.3M	2.5M	996k	583k	498k
"manufacturing"	"CAD"	"spare parts"	"3D printing"	"datasheet"

Also present: CNC machining (150k) · injection molding (120k) · industrial automation (562k) · robotics (788k) · PLC (782k) — a domain corpus you can filter out of the OWI.

Keyword matches over one month, English-leaning — a **candidate pool** before curation, dedup, and licensing.

Manufacturing is **just one example** — the OWI holds rich, filterable corpora for *any* domain (law, health, science, finance, engineering, …), **millions of documents per month**.

Document counts from the Open Web Index API (api.ourrs), one-month corpus, queried 2026-05-31.

Structured Web Data

OWI parses **schema.org / Microdata** and classifies pages — including **forums and Q&A** — into **453.4M typed entities**, with three kinds of value:

~83M Deeper knowledge

articles (79.4M) · scholarly (897k) · books (1.7M) · datasets · courses · medical

→ **training & RAG**

~39M Behavioural

blog posts (18.2M) · questions (17.5M) · Q&A pages (717k) · reviews · social posts · forums

→ **instruction & evaluation**

~120M Search & agents

products (91M) · persons (10M) · organisations (6.2M) · events (5.2M) · search actions (3.7M)

→ **agentic search & affordances**

The same crawl yields **knowledge to learn from, behaviour to learn from, and entities to act on**.

Entity counts by schema.org primary_type, Open Web Index API (api.ourrs), corpus 2026-05.

Search becomes the memory layer

What lives in the weights, and what stays retrievable?

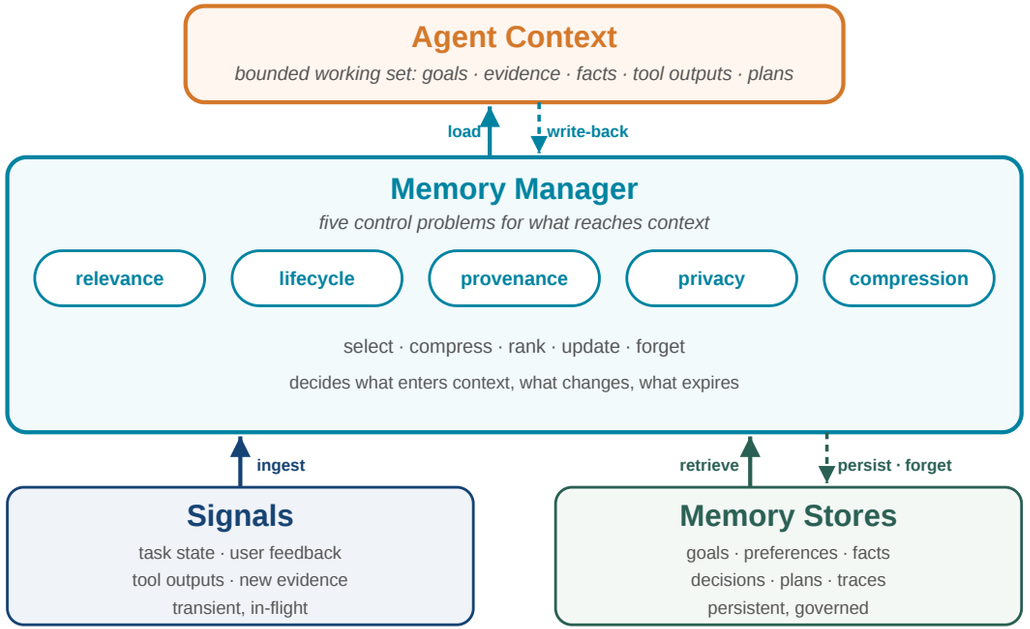
Search as Memory: Parametric vs. Non-Parametric

Retrieval-Augmented Generation (RAG)

Search for context relevant documents.
Usually as Vector search, to cover linguistic and semantic diversity.

RAG introduces a "memory" hierarchy:

- **parametric:** model capabilties & internal knowledge
- **non-parametric:** external knowledge



RAG is essential for freshness, domain specific adaption and data privacy, but requires scalable and efficient **search infrastructures** in an AI system

Lewis et al. 2020, Retrieval-Augmented Generation (arXiv:2005.11401); parametric vs. non-parametric memory, Mallen et al. 2023

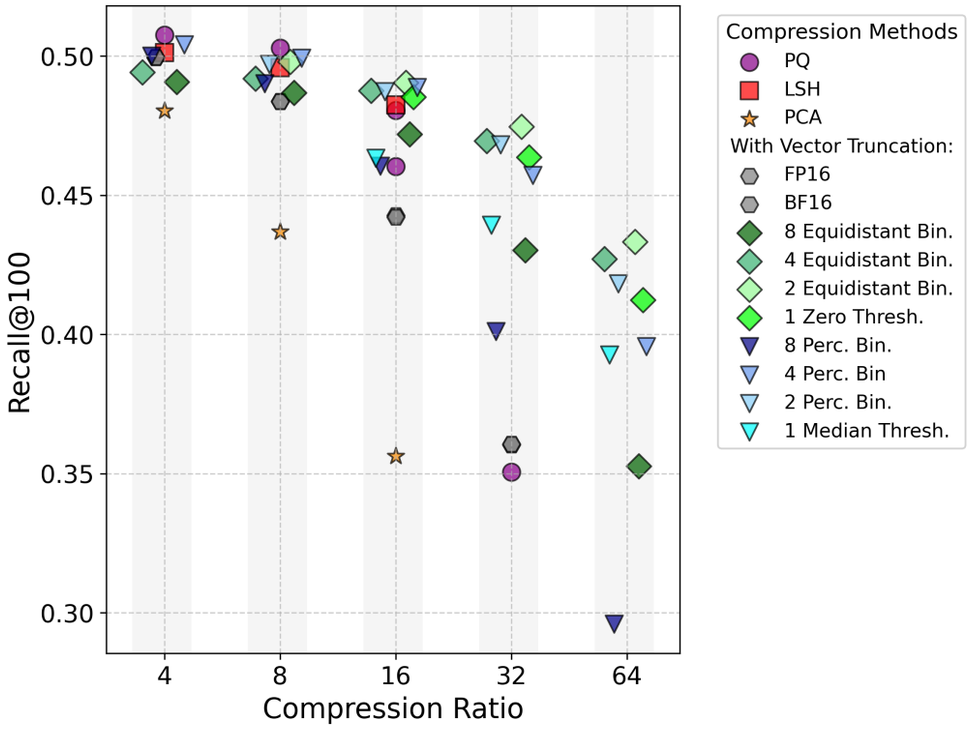
Web-Scale Vector Memory Has a Cost

The Open Web Index provides **embeddings** for retrieval on selected web documents, but increases compute and storage significantly.

Why compression matters

- Compute is managable / scalable
- But embeddings are **expensive to store and serve** *($O(n)$ for exact queries)*
- compression trades index size against retrieval quality
- Quantisation wins: results hold recall while compressing **16–32×**

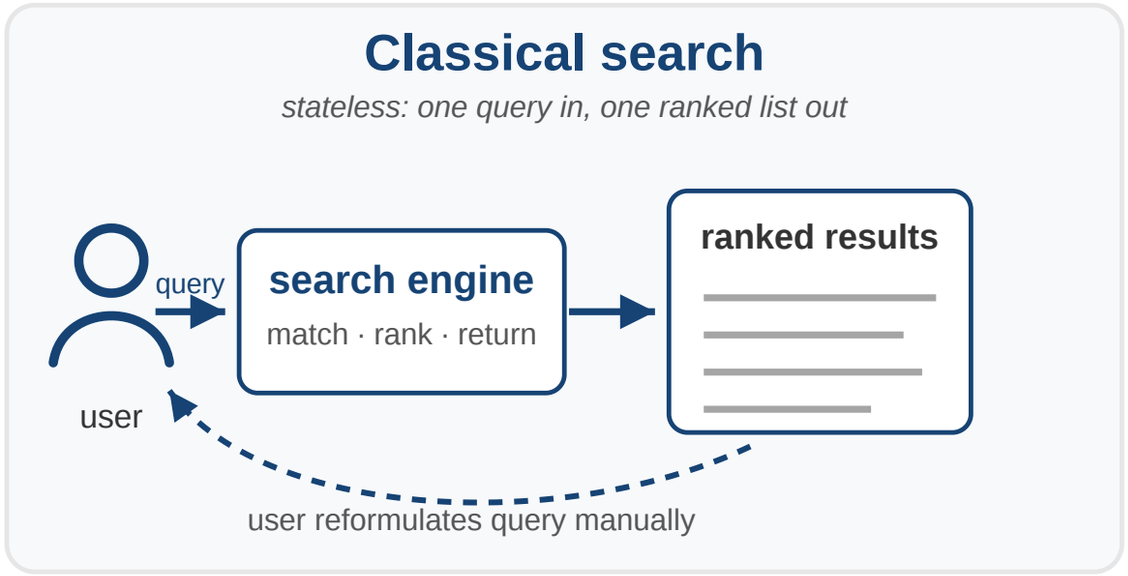
> Compression essential for **scaling up search**.

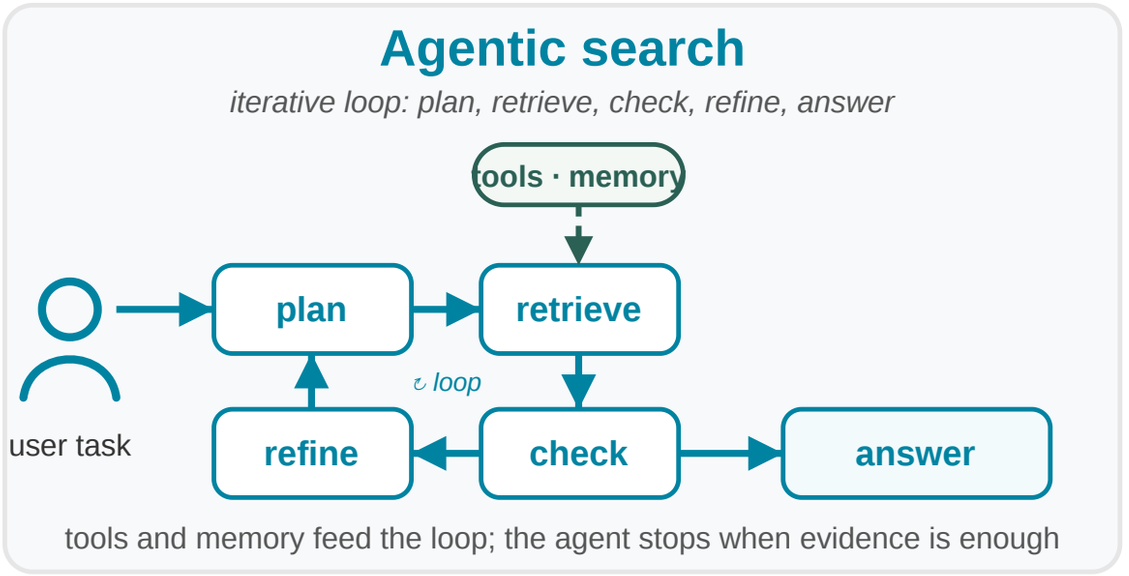


From retrieval to agents

Retrieval moves into a control loop — and needs actions and traces.

Agentic Search: Retrieval in a Control Loop

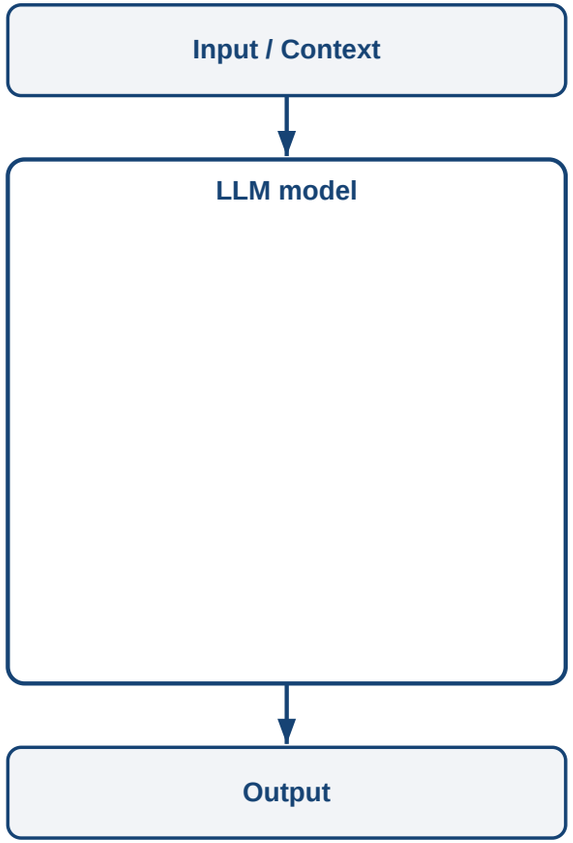




Agents do not just **consume** search results — they **control** retrieval: plan, inspect evidence, call tools, reformulate, synthesize with provenance.

Related paradigms: ReAct (arXiv:2210.03629); Toolformer (arXiv:2302.04761); Self-RAG — adaptive, self-reflective retrieval (arXiv:2310.11511).

A System View of Agentic Search

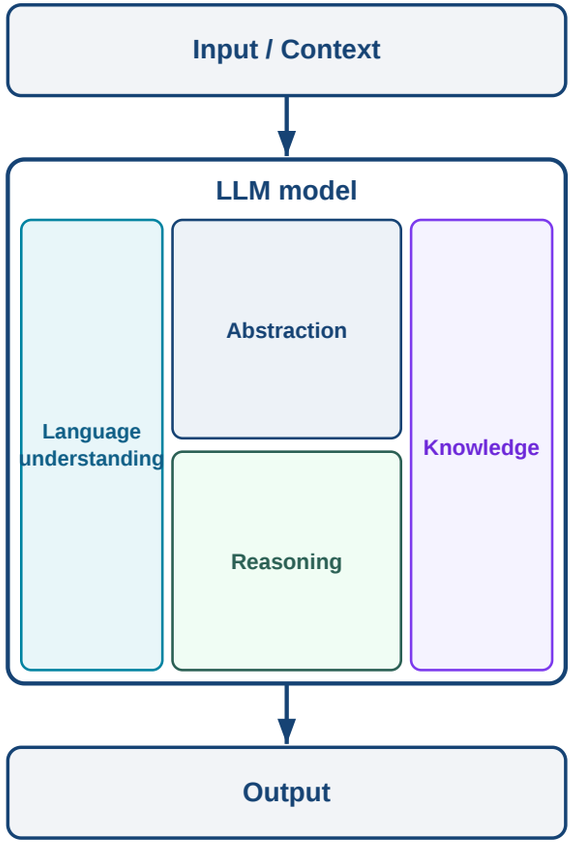


How much knowledge must sit in **weights** vs. **external memory**? Externalising it → **smaller model + fresher, controllable knowledge** — trading latency & more loops for

flexibility, control. *Where to invest resources?*

Memory roles after CoALA: working · episodic · semantic · procedural (Sumers et al. 2023, arXiv:2309.02427).

A System View of Agentic Search

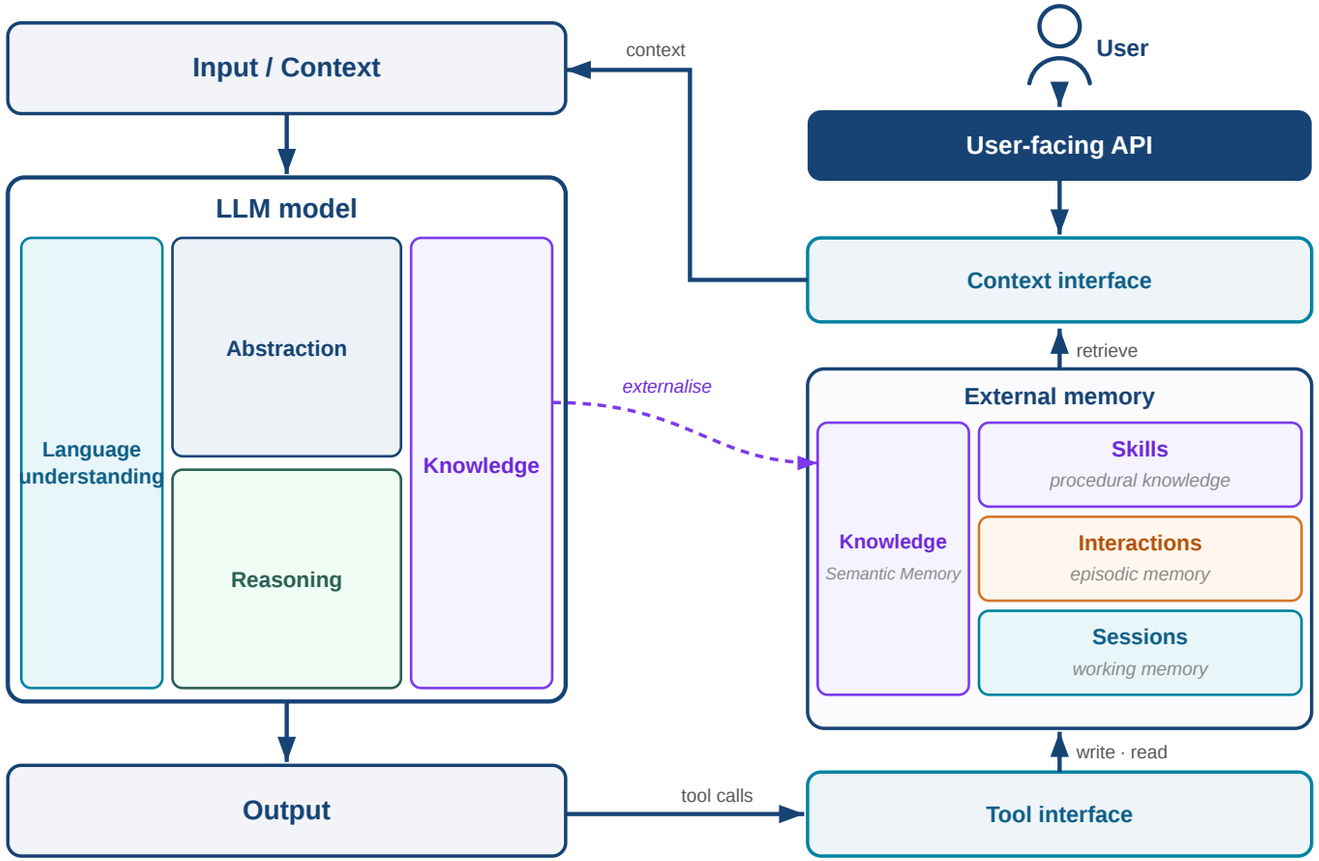


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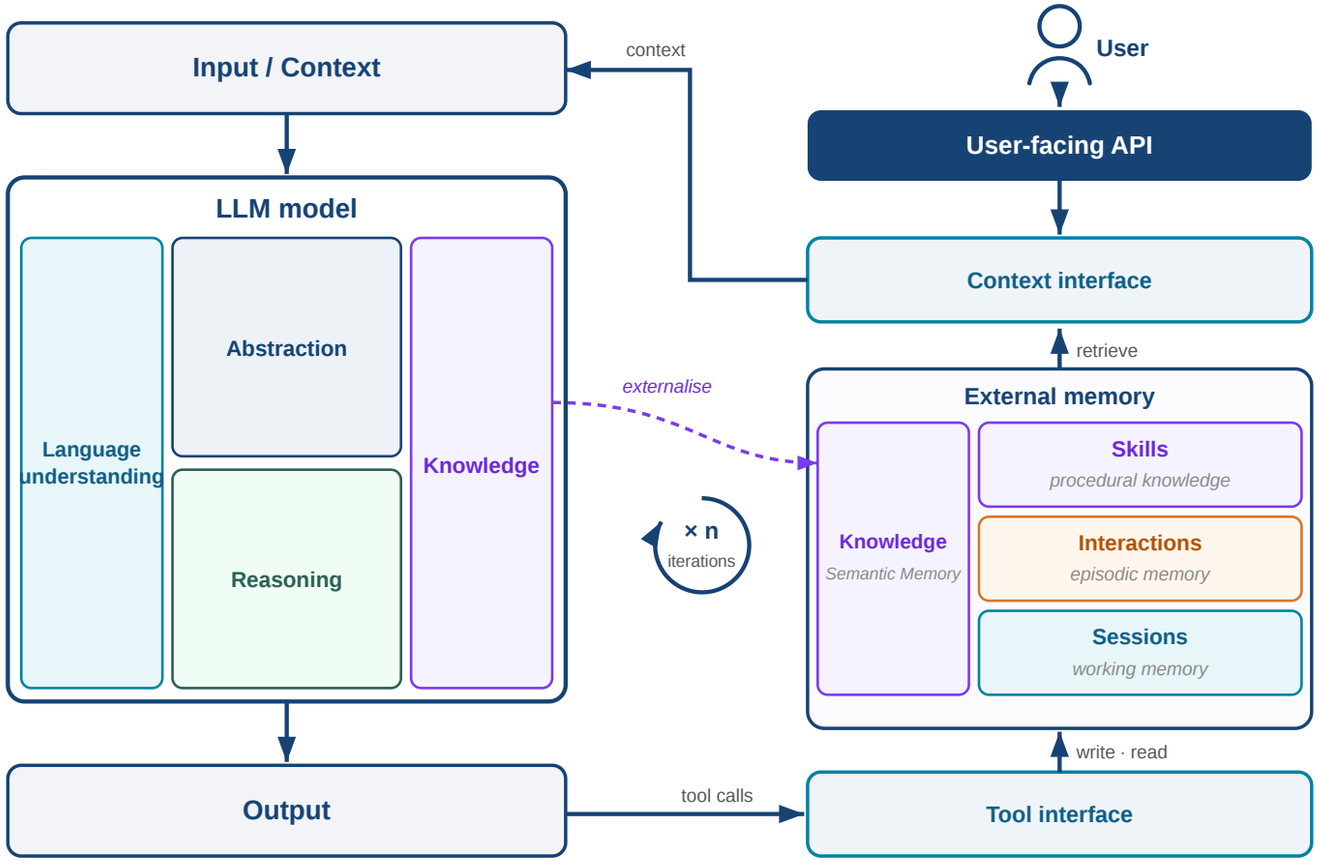


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Structured Web Data Becomes Agent Affordance

ACTION & APPLICATION ENTITIES IN THE OPEN WEB INDEX

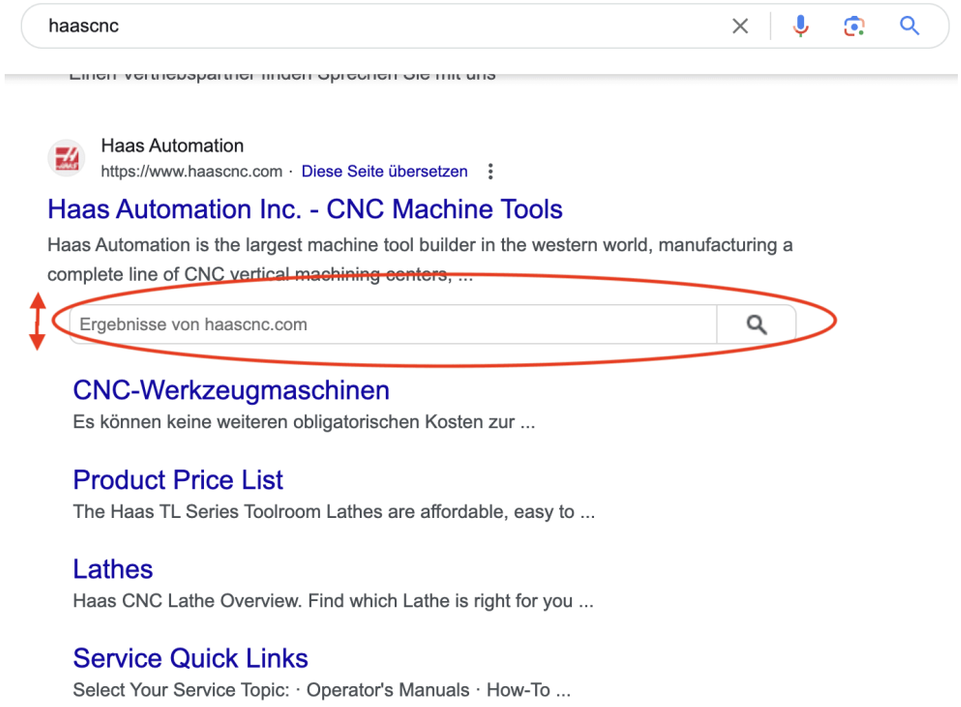
3.65M
SearchAction

909k
SoftwareApplication

114k
WebApplication

94k
MobileApplication

Agents need **discoverable actions** — what can be searched, called, installed, or configured. The index is a **candidate generator**, not blind trust.



*Sitelinks searchbox = schema.org **SearchAction**: agents can call a site's own search. Source: metayer.de*

Entity counts by schema.org primary_type, Open Web Index API (api.ourrs), corpus 2026-05.

Beyond Web Data: Traces & Evaluation

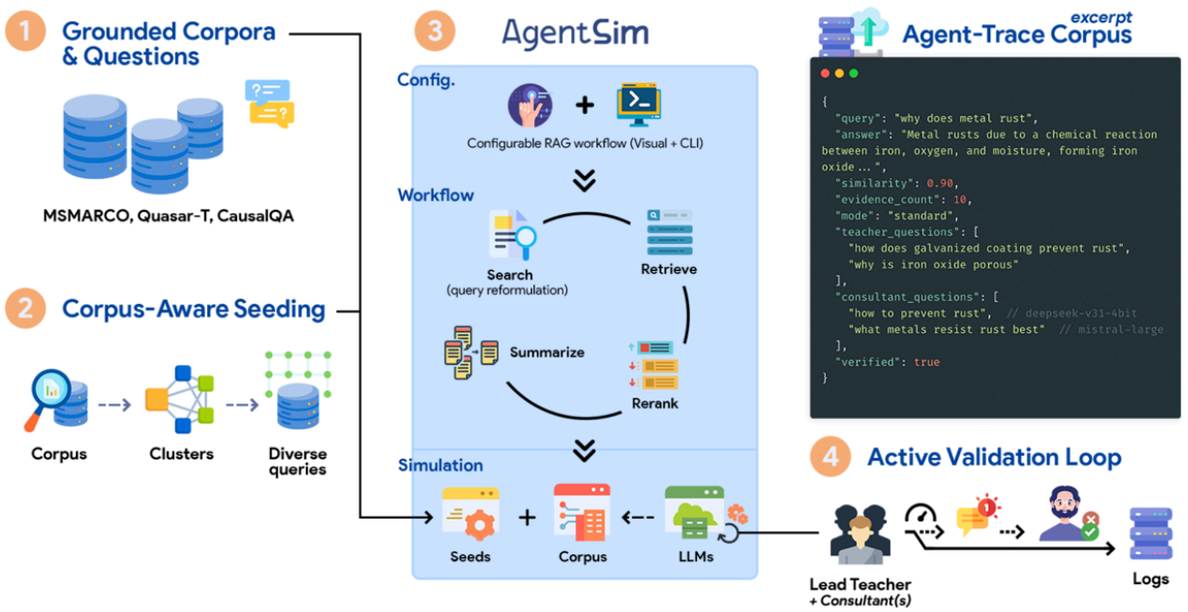
OPEN CHALLENGES IN AGENTIC SEARCH — OUR WORK

Synthetically creating agent traces

AgentSim — simulates RAG agents over document collections to generate synthetic, **document-grounded** reasoning traces

- **103k+** verifiable steps (Agent-Trace Corpus)
- 100% grounding rate on substantive answers.
- Agent-trace synthetic data generating is both **GPU compute heavy** and requires **search systems**.

AgentSim, Agent-Trace Corpus · arXiv:2604.26653



Comparing to simulated human users

UXSim — grounds an adaptive LLM agent with classical simulators → accurate, **explainable** user-search simulation without costly user studies.



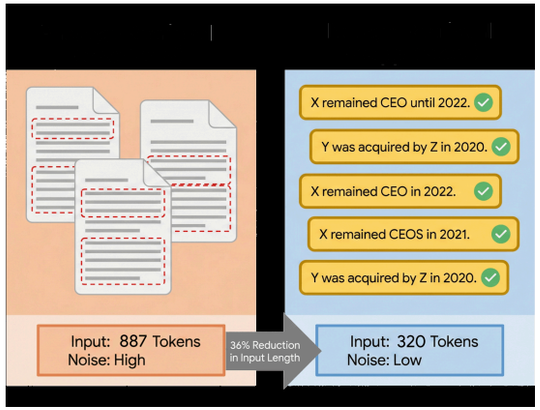
Beyond Web Data: Knowledge & Interaction

OPEN CHALLENGES IN AGENTIC SEARCH — OUR WORK

How to structure knowledge so agents find it?

NuggetIndex — distil documents into atomic **"nuggets"**: 887 → 320 tokens, far less noise, facts kept for grounding.

Own work — nugget-level indexing for agent knowledge



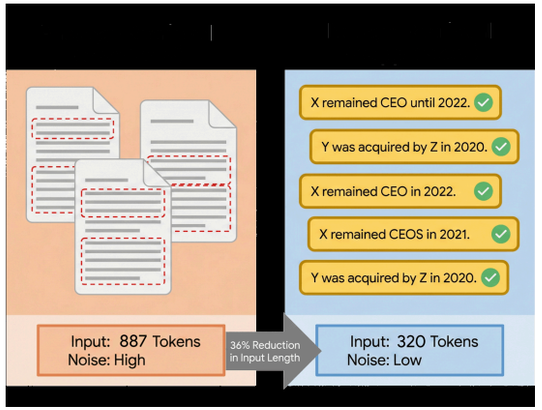
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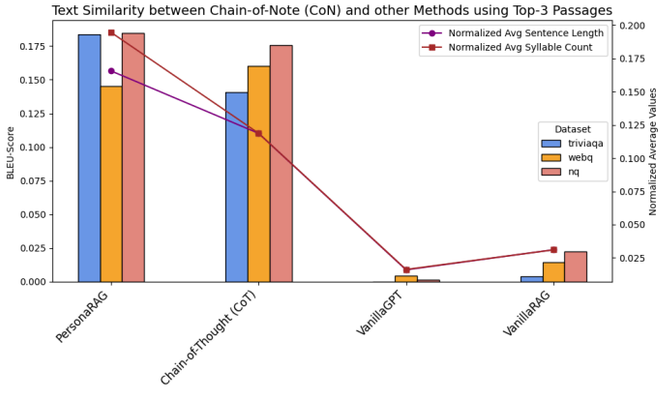
Own work — nugget-level indexing for agent knowledge



How should retrieval adapt to the user?

PersonaRAG — user-centric agents personalise retrieval → higher answer quality (BLEU) than CoT and vanilla RAG across TriviaQA, WebQ, NQ.

Zerhoudi & Granitzer 2024 · arXiv:2407.09394



Conclusion

Tokens pretraining → **Examples** fine-tuning → **Passages** RAG memory → **Actions** tools & APIs → **Traces** behaviour & feedback

What we covered

- The web is an open, reusable **data layer** — OpenWebSearch delivers **data + an index**
- **Filter** it for training & fine-tuning; **serve** it as RAG memory & agentic search
- Agentic search needs more than data: **interaction, knowledge structure, behaviour, evaluation**
- Developing Agentic Search is not only GPU-intensive, it requires complex systems.

Let's cooperate

We are **looking forward to collaborations** on open web data, retrieval memory, and agentic search — research projects, datasets, and use-cases all welcome.

→ openwebsearch.eu · Chair of Data Science, University of Passau

Better models and better data matter **only when they become systems people use.**

References

OpenWebSearch.eu: https://openwebsearch.eu/

TPC26: https://tpc26.org/ · LRZ goes TPC26: https://www.lrz.de/en/latest-news/events/detail/lrz-goes-tpc26

Lewis et al. 2020, Retrieval-Augmented Generation: https://arxiv.org/abs/2005.11401

CoRECT 2025, Controlled Retrieval Evaluation of Compression Techniques: https://arxiv.org/abs/2510.19340

Kandpal et al. 2023, LLMs Struggle to Learn Long-Tail Knowledge: https://arxiv.org/abs/2211.08411

Mallen et al. 2023, When Not to Trust Language Models: https://arxiv.org/abs/2212.10511

Yao et al. 2022, ReAct: https://arxiv.org/abs/2210.03629

Schick et al. 2023, Toolformer: https://arxiv.org/abs/2302.04761

Asai et al. 2023, Self-RAG: https://arxiv.org/abs/2310.11511

Sumers et al. 2023, Cognitive Architectures for Language Agents (CoALA): https://arxiv.org/abs/2309.02427

Zerhoudi & Granitzer 2024, PersonaRAG: https://arxiv.org/abs/2407.09394

AgentSim 2026, Agent-Trace Corpus: https://arxiv.org/abs/2604.26653

Zerhoudi & Granitzer 2025, UXSim — Hybrid User Search Simulation: https://arxiv.org/abs/2602.24241